

Day 10 – Customized EC2 Creation with VPC Concepts

Topics Covered

- What is VPC
 - CIDR & IP Range
 - Class A IP Range
 - Public & Private Subnets
 - Internet Gateway (IGW)
 - Route Tables
 - Security Groups
 - Customized EC2 Launch inside Custom VPC
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1. What is VPC?

A **VPC (Virtual Private Cloud)** is a logically isolated network in AWS where we create and manage resources like:

- EC2
- RDS
- Load Balancer
- EKS
- Lambda (inside VPC)

It is similar to a private data center network in cloud.

2. Understanding CIDR Range

CIDR notation decides how many IP addresses are available.

Example:

10.0.0.0/16

Where:

- 10.0.0.0 → Network address
- /16 → Subnet mask

Total IPs:

65536 IPs

3. Class A IP Range

Class A private IP range:

10.0.0.0 – 10.255.255.255

Example VPC Range:

10.0.0.0/16

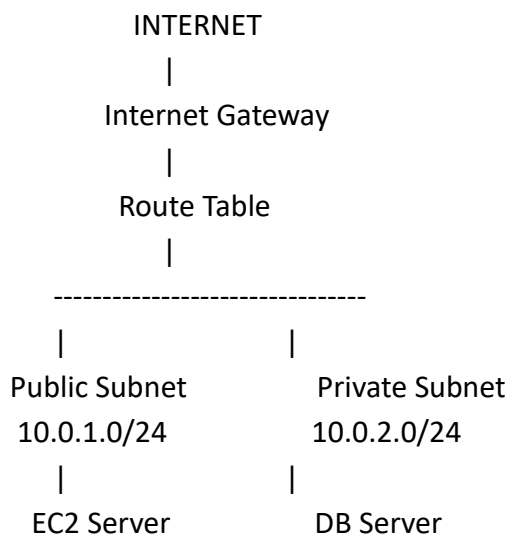
Subnet Examples:

Subnet	CIDR
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Public Subnet	10.0.1.0/24
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Private Subnet	10.0.2.0/24
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4. Architecture Diagram



5. Step-by-Step Custom VPC Creation

Step 1 – Create VPC

Go to:

AWS Console → VPC → Create VPC

Details:

Option	Value
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Name	custom-vpc
------	------------

IPv4 CIDR	10.0.0.0/16
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Click:

Create VPC

6. Create Public Subnet

Go to:

VPC → Subnets → Create Subnet

Details:

Option	Value
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Name	public-subnet
------	---------------

CIDR	10.0.1.0/24
------	-------------

AZ	ap-south-1a
----	-------------

7. Create Private Subnet

Option	Value
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Name	private-subnet
------	----------------

CIDR	10.0.2.0/24
------	-------------

8. Create Internet Gateway (IGW)

Go to:

VPC → Internet Gateways → Create

Name:

custom-igw

After creation:

Attach to VPC

Purpose:

- Provides internet access to public subnet resources.
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9. Create Route Table

Go to:

VPC → Route Tables → Create

Name:

public-route-table

Add Route:

Destination Target

0.0.0.0/0 IGW

Meaning:

- All internet traffic goes through IGW.

Associate:

public-subnet

10. Enable Auto Public IP

Go to:

Subnet → Actions → Edit subnet settings

Enable:

Auto-assign public IPv4

Without this:

- EC2 will not get public IP.
- Internet access will fail.

11. Create Security Group

Go to:

EC2 → Security Groups → Create

Name:

web-sg

Inbound Rules:

Type	Port	Source
SSH	22	My IP
HTTP	80	Anywhere
HTTPS	443	Anywhere
Custom TCP 8080 Anywhere		

Purpose:

- Acts like firewall for EC2.

12. Launch EC2 in Custom VPC

Go to:

EC2 → Launch Instance

Choose:

- Ubuntu/Linux AMI
- t2.micro

Network Settings:

Option	Value
VPC	custom-vpc
Subnet	public-subnet
Public IP	Enable
Security Group	web-sg
Launch instance.	

13. Verify Connectivity

SSH into server:

```
ssh -i key.pem ubuntu@PUBLIC-IP
```

Check internet:

```
ping google.com
```

Install web server:

```
sudo apt update
```

```
sudo apt install nginx -y
```

Access:

```
http://PUBLIC-IP
```

14. Real-Time Interview Questions

Q1. Why do we use VPC?

To isolate AWS resources securely.

Q2. Difference between Public & Private Subnet?

Public	Private
Has internet access	No direct internet

Public	Private
Uses IGW	Uses NAT Gateway
Web servers	Databases

Q3. What is IGW?

Internet Gateway provides internet connectivity to VPC.

Q4. What is Route Table?

Controls traffic routing inside VPC.

Q5. Difference between NACL & Security Group?

Security Group NACL

Instance level	Subnet level
Stateful	Stateless
Allow rules only	Allow & Deny

15. Important Concepts Summary

Component	Purpose
VPC	Private network
Subnet	Division of VPC
IGW	Internet access
Route Table	Traffic routing
Security Group	Firewall
EC2	Virtual server

Day 10 Task

Practical Task

Create:

- 1 Custom VPC
- 1 Public Subnet
- 1 Private Subnet
- 1 IGW
- 1 Route Table
- 1 Security Group
- 1 EC2 Instance

Then:

- Install NGINX
- Access webpage from browser